

Machine_Learning_Engineer — Step 2

Master Statistics & Probability

Detailed Learning Notes • CareerCompass

1. Learning Objective

Develop the statistical reasoning needed to summarize data, quantify uncertainty, design experiments and evaluate whether a model result is meaningful.

2. Core Concepts — Detailed Breakdown

Descriptive statistics

Use mean, median, mode, range, variance, standard deviation and percentiles to summarize distributions. Learn when the median is more robust than the mean, especially with skewed data and outliers.

Distributions and sampling

Understand normal, binomial and other common distributions conceptually. Learn population versus sample, sampling bias, sampling distributions and why repeated samples produce variability.

Inference

Understand confidence intervals, hypothesis tests, null and alternative hypotheses, p-values and practical versus statistical significance. A small p-value is not automatically a large or useful effect.

Relationships between variables

Study covariance, correlation and simple regression. Learn how confounding variables can create misleading relationships and why randomized experiments are stronger for causal claims.

Probability for ML

Understand conditional probability, Bayes' rule, independence and expected value. These ideas support probabilistic models, classification metrics, uncertainty estimation and reasoning about rare events.

3. Recommended Learning Workflow

- Start with descriptive statistics and distributions.
- Practice probability with tables and simple simulations.
- Perform confidence-interval and hypothesis-testing exercises.
- Analyze correlation and regression on a small dataset.
- Write conclusions in plain language, separating evidence from assumptions.

5. Hands-on Project / Practice

Run a statistical investigation on a public dataset. Form one question before looking at the result, define the population/sample, calculate descriptive statistics, visualize the data, perform one suitable inferential test, and write a conclusion that explicitly states limitations and practical significance.

6. Common Mistakes & Pitfalls

- Choosing a statistical test only because software offers it.

- Treating $p < 0.05$ as proof of a business effect.
- Ignoring sampling bias or dependence between observations.
- Reporting a correlation as causal evidence.
- Using an inappropriate distributional assumption without checking the data.

7. Completion Checklist

- I can explain population vs sample.
- I can interpret mean, variance, standard deviation and percentiles.
- I can explain confidence intervals and p-values.
- I can distinguish statistical significance from practical significance.
- I can reason about conditional probability and independence.

8. Interview / Assessment Questions

- What does a p-value actually measure?
- Why does sample size affect statistical significance?
- When is median preferable to mean?
- What is conditional probability?
- Why does correlation not establish causation?

9. Recommended References

Khan Academy Statistics & Probability: <https://www.khanacademy.org/statistics-probability>

Machine Learning in 2024 — freeCodeCamp: <https://www.youtube.com/watch?v=bmmQA8A-yUA>

Data Science Tutorial — freeCodeCamp: <https://www.youtube.com/watch?v=ua-CiDNNj30>

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.